Causal Self-Consistency of the Speed of Light in Fractal S^3 Geometry

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Abstract. This work does not derive the numerical value of the speed of light from independent first principles. Instead, it formulates a *causal self-consistency test* for fractal S^3 volume–density evolution. The speed of light c is taken as the empirically observed invariant causal speed, and the question posed is whether the proposed S^3 hierarchy admits a closed, self-consistent volume–density transition under that invariant boundary condition. A golden-ratio scaling $R_n = R_0\varphi^n$ is used as a structural hierarchy, while any Planck-scale normalisation is treated only as an illustrative calibration, not as a derivation. The resulting identity $c = c$ is interpreted as a *closure condition*: it shows that the assumed geometric and dynamical relations remain internally consistent under the empirical causal speed, rather than providing an independent prediction of c.

Keywords: empirical invariant c; causal boundary; self-consistency test; closure condition; fractal S^3 hierarchy; structural ansatz; conceptual consistency framework.

1. Introduction

In standard physics the speed of light

$$c = 299\,792\,458\ \mathrm{m\,s^{-1}}$$

is an empirically measured invariant, enshrined in the SI definition of the metre. Its invariance is a postulate of special relativity, established through measurement and not derived from a more primitive geometric principle.

The present work adopts a different, more modest aim. We **do not** claim to derive c from deeper constants or abstract first principles. Instead, we **test whether the proposed fractal S^3 hierarchy is internally consistent with the empirically observed invariant causal speed** c. Specifically:

1. c is accepted as the empirically measured invariant causal speed. No attempt is made to predict its numerical value independently.

2. The model examines whether the fractal S^3 volume–density transition admits a self-consistent *fixed-point condition* when the causal boundary propagates at c.

3. The closure identity $c = c$ that arises from substitution is interpreted as a fixed-point / closure condition, confirming internal geometric coherence rather than constituting a derivation.

Because an observer inside a single causal universe cannot measure light from "outside" the universe, c is operationally accessed only through internal empirical procedures. The present model therefore treats c as *empirical input* and tests whether the proposed S^3 structure closes consistently under this input.

2. Fractal S^3 Hierarchy

2.1 Golden-ratio scaling as structural ansatz

The fractal S^3 manifold is parameterised by level radii

$$\boxed{R_n = R_0\,\varphi^n,} \qquad n \in \mathbb{Z},$$

where $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ is the golden ratio and $n = 0$ marks the recombination epoch, at which photons decouple from matter and propagate freely.

Status of the ansatz. The golden-ratio hierarchy is used as a minimal self-similar scaling structure. Its mathematical motivation is the Perron–Frobenius dominance of the minimal positive second-order recursion, but its physical realisation in the present light-boundary model remains a *structural ansatz*. We do not assert that φ is the uniquely valid scaling ratio for an S^3 geometry; we adopt it as a discrete self-similar hierarchy and test the resulting consistency.

2.2 Optional Planck-scale calibration

A Planck-scale seed may be used as a reference normalisation,

$$R_0 = l_{\mathrm{P}}\,\varphi^N, \qquad l_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35}\ \mathrm{m},$$

connecting the quantum scale to a cosmological boundary scale through the golden-ratio hierarchy. For the recombination-horizon scale $R_0 \approx 2.536 \times 10^8$ m, this gives

$$N = \frac{\ln(R_0/l_{\mathrm{P}})}{\ln\varphi} \approx 206.8 \implies N = 207.$$

Calibration note. The value $N \approx 207$ is used only as one illustrative calibration connecting a Planck-scale seed to a recombination-scale boundary. It is *not* required for the logical structure of the closure condition developed in Section 5, and it does not constitute a derivation of c or of any cosmological observable.

3. Volume, Density, and Mass Normalisation

3.1 Global S^3 volume

For a three-sphere S^3 of radius R, the exact global volume is

$$V_{S^3}(R) = 2\pi^2 R^3.$$

This is the geometrically correct definition for the compact, simply connected, boundaryless S^3 manifold used in this framework.

3.2 Local Euclidean approximation

In the regime of small hyperspherical coordinate ($\chi \ll 1$), the S^3 volume reduces locally to the Euclidean approximation

$$V_{\text{local}}(R) \approx \tfrac{4}{3}\pi R^3.$$

Scope of the approximation. The local Euclidean form is employed below only for illustrative dynamical scaling in the $\chi \ll 1$ regime. It does not replace the global S^3 volume definition, and a fully covariant treatment would use V_{S^3} throughout.

3.3 Mass as a closure normalisation parameter

The matter density at the zeroth level is parameterised as

$$\rho_0 = \frac{M_0}{\frac{4}{3}\pi R_0^3}.$$

The mass M_0 is a *normalisation parameter chosen to test closure of the dynamical relation under the empirical causal speed* c. It is not derived from first principles, and its value is determined self-consistently at the closure step (Section 5).

4. Causal Boundary Dynamics

4.1 Boundary velocity

The rate at which the fractal boundary advances between consecutive levels is

$$c_n = \frac{R_{n+1} - R_n}{\Delta t_n} = \frac{R_0 \varphi^n (\varphi - 1)}{\Delta t_n}.$$

At the zeroth level ($n = 0$):

$$c = \frac{R_0(\varphi - 1)}{\Delta t_0}.$$

We *posit* that this boundary velocity equals the empirically measured c, and subsequently test whether this assumption is maintained self-consistently.

4.2 Effective expansion rate

Using the Friedmann-type effective Hubble parameter,

$$H_0 = \sqrt{\frac{8\pi G \rho_0}{3}} = \sqrt{\frac{2GM_0}{R_0^3}},$$

the causal boundary velocity becomes

$$c = H_0 \, R_0 = \sqrt{\frac{2GM_0}{R_0}}.$$

Incorporating the fractal correction $1/\varphi$ (arising from the golden-ratio step between adjacent levels):

$$c = \sqrt{\frac{2GM_0}{R_0}} \cdot \frac{1}{\varphi}. \tag{1}$$

5. Closure Condition

Solving equation (1) for the mass normalisation that makes the relation consistent with the empirical invariant c gives:

$$M_0 = \frac{c^2\, R_0\, \varphi^2}{2G}.$$

Substituting back into (1):

$$c_{\text{out}} = \sqrt{\frac{2G}{R_0} \cdot \frac{c^2 R_0 \varphi^2}{2G}} \cdot \frac{1}{\varphi} = \sqrt{c^2\varphi^2} \cdot \frac{1}{\varphi} = c\varphi \cdot \frac{1}{\varphi} = c. \tag{2}$$

Closure condition. The substitution yields the identity $c = c$. This identity is expected by construction once M_0 is chosen as the closure normalisation.

The point is *not* to predict c independently. The point is to show that **the proposed dynamical chain admits a self-consistent solution when the causal boundary speed is the empirical invariant** c: the assumed fractal S^3 scaling, density–volume relation, and causal-boundary dynamics form a closed, coherent system under that boundary condition. The result is therefore a *fixed-point condition for causal self-consistency*, not an independent prediction of the speed of light.

The structure of the constraint is the following. The golden-ratio hierarchy, Planck-scale calibration, and Friedmann-type dynamics together admit a self-consistent solution when the boundary propagation speed equals c. Modifying the scaling ratio, the calibration level N, or the density–volume relation changes the algebraic structure and may or may not admit a self-consistent closure. Within the present model calibration, c appears as the consistent propagation speed — the *fixed point of causal self-consistency*.

6. Analogy with Maxwell Electrodynamics

In Maxwell's electrodynamics, the speed of light appears as

$$c = \frac{1}{\sqrt{\varepsilon_0\, \mu_0}},$$

relating the propagation speed of electromagnetic waves to the electric permittivity ε_0 and magnetic permeability μ_0 of the vacuum. This relation does not derive c ex nihilo. It relates c to independently measured electromagnetic constants and shows that the internal structure of Maxwell's equations singles out a characteristic invariant speed.

Similarly, the present framework does not derive c from nothing. It tests whether a fractal S^3 volume–density hierarchy admits a causal closure when the boundary speed is the empirical invariant c. In both cases, c appears not as an arbitrary free parameter, but as *the speed required for internal consistency of the theoretical structure*.

Epistemological note. This analogy is intended as an epistemological comparison only. We do not claim that the present framework achieves the same empirical scope or predictive status as Maxwell's theory. The analogy illustrates a shared logical structure: in both cases, c is singled out by an internal consistency requirement rather than introduced as a free parameter.

7. Causal Self-Consistency Principle

Causal Self-Consistency Principle

A compact, simply connected, boundaryless S^3 geometry undergoing fractal volume–density evolution governed by $R_n = R_0\varphi^n$ admits a coherent causal-boundary description when the boundary propagation speed is the empirical invariant c. Within the present model calibration, the golden-ratio hierarchy, Planck-scale normalisation, and Friedmann-type dynamics together select c as the consistent propagation speed — the fixed point of causal self-consistency.

This principle is understood as a *necessary condition for internal coherence within the stated model*. It is not asserted as a uniqueness theorem without further derivation: asserting "if and only if" would require a separate proof that no other boundary speed admits any reparametrisation of the model that also closes consistently. The present paper establishes the fixed-point character of c, not its uniqueness in a strong mathematical sense.

8. Numerical Illustration

We verify that the closure condition (2) is numerically realised for standard values of the physical constants, using the $N = 207$ illustrative calibration.

Parameter	**Symbol**	**Value**
Planck length	l_{P}	$1.616 \times 10^{-35}\ \mathrm{m}$
Golden ratio	φ	1.618
Base scale (calibration)	$R_0 = l_{\mathrm{P}}\varphi^{207}$	$2.536 \times 10^{8}\ \mathrm{m}$
Gravitational constant	G	$6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$
Speed of light *(empirical input)*	c	$2.998 \times 10^{8}\ \mathrm{m\,s^{-1}}$

Closure normalisation:

$$M_0 = \frac{c^2 R_0 \varphi^2}{2G} = \frac{(2.998 \times 10^8)^2 \cdot 2.536 \times 10^8 \cdot (1.618)^2}{2 \times 6.674 \times 10^{-11}} \approx 4.47 \times 10^{35}\ \mathrm{kg}.$$

Recovered causal boundary speed:

$$c_{\mathrm{out}} = \sqrt{\frac{2GM_0}{R_0}} \cdot \frac{1}{\varphi} = \sqrt{\frac{2 \times 6.674 \times 10^{-11} \times 4.47 \times 10^{35}}{2.536 \times 10^8}} \cdot \frac{1}{1.618} \approx 2.998 \times 10^8\ \mathrm{m\,s^{-1}}.$$

This numerical agreement reflects the *imposed closure condition* and does not constitute an independent prediction of c. The recovered value equals the input because M_0 was defined precisely to enforce this agreement. The numerical exercise demonstrates that the closure is algebraically and numerically coherent, not that c has been derived from more primitive quantities.

9. Conclusion

The results of this paper may be summarised as follows.

1. **No derivation of c.** This work does not derive the numerical value of the speed of light from independent first principles. c is treated throughout as the empirically observed invariant causal speed.

2. **Closure structure.** The fractal S^3 hierarchy $R_n = R_0\varphi^n$, the density–volume relation, and the causal-boundary dynamics together form a self-consistent closure structure when the boundary propagation speed is set to the empirical invariant c. The identity $c = c$ arising from substitution is a *fixed-point condition for causal self-consistency*.

3. **Epistemological analogy.** The logical structure parallels Maxwell's relation $c = 1/\sqrt{\varepsilon_0\mu_0}$: in both cases, c is not an arbitrary free parameter but the speed required for internal consistency of the theoretical framework.

4. **Optional calibration.** The Planck-scale normalisation $R_0 = l_{\mathrm{P}}\varphi^N$ with $N \approx 207$ is an illustrative calibration connecting quantum and cosmological scales through the golden-ratio hierarchy. It is not a fundamental derivation and is not required for the logical structure of the closure condition.

5. **Scope.** This paper is a conceptual consistency study in fractal S^3 geometry. Its contribution is to show that the proposed geometric and dynamical framework is internally coherent under the empirical causal speed c.

Thus, c should not be understood here as a number derived from more primitive constants, but as the **empirical invariant causal speed that makes the proposed fractal S^3 volume–density evolution close consistently**. The result is a fixed-point condition for causal self-consistency — a conceptual consistency framework, not an independent prediction of the speed of light.

Namaste.

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